Simple model of black hole physics

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In this problem we will take a look at black holes - one of the most fascinating objects known to mankind.

Introduction/Brief Theory Review

A black hole in astrophysics is a sufficiently massive object, the gravitational attraction of which is so strong that even light cannot leave her immediate surroundings. Several types of black hole are believed to exist, but in this problem we will only look at the types which mark the end of stellar evolution.

The equilibrium of ordinary stars is supported by thermonuclear reactions within the star,which opposes gravitational compression. After the thermonuclear fuel reserves are exhausted, the temperature of the star drops sharply, rapid gravitational compression (collapse) occurs, as a result of which the electrons of the matter are pressed into the protons and eventually neutral matter of density $\rho = 1,1 \times 10^{18}\,\mathrm{kg/m^3}$ is formed. For the entirety of the problem assume that the star remains spherically symmetric and homogeneous.

Problems

- Calculate the minimum mass M_{min} of a black hole.

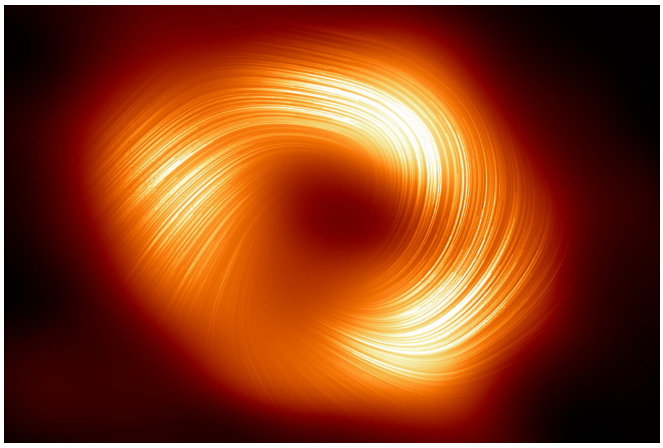

Figure 1: *Image of the first photographed black hole in the elliptical galaxy M87.*

Mechanics of a black hole

Consider a star X, which immediately before gravitational collapse has a mass $M = 3.00 \times 10^{32}$kg and a radius of $R_0 = 7.10 \times 10^8$m which rotates around its axis with a period of $T = 2.00 days$ and has a magnetic field near the equator with an induction of $B_0 = 2.30 \times 10^{-2} T$.

- Calculate the radius of the resulting black hole R
- Calculate the period of the black hole's rotation around its axis T.
- Calculate the location of the event horizon of the resulting black hole, i.e. the distance R_g

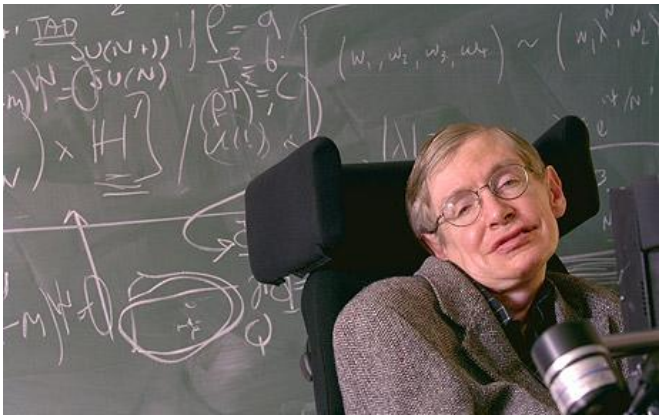

Figure 2: *Stephen Hawking (1942 - 2018).*

from which it is still theoretically possible to obtain some information about the processes occurring there.

Magnetic influence

When a star collapses,the magnetic field lines are frozen into the matter, that is, they are compressed along with it.

- Calculate the magnitude of the magnetic induction B near the equator of the star at the moment when its radius was $2R_g$.
- Calculate the work of gravitational forces A performed on the matter of the star X during its collapse from the initial state to a radius of $2R_g$.

Hawking radiation

Stephen Hawking suggested in 1975 that a black hole, despite its strong gravitational field, should radiate. This radiation should consist mainly of photons, but it has not yet been detected experimentally. As a result of the radiation, the mass of the black hole M should decrease according to the law

$$\frac{dM}{dt} = -\frac{\hbar c^5}{15360\pi G^2 M^2}$$

- Calculate the lifetime τ from the moment of formation of a black hole until its transformation into an observable object.

The transformation of an initial normal star into a black hole is possible only if its initial period of rotation around its axis is not less than some minimum value T_{min}.

- Calculate the ratio $\frac{T_0}{Tmin}$ for the above parameters of the star X.

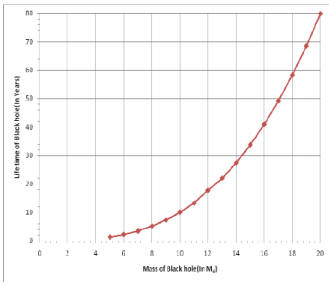

Figure 3: *Plot of mass vs life expectancy of a black hole*

Fundamental constants

Constant	Value	Units
g	9.8	N/kg
G	6.67×10^{-11}	N·m^2/kg^2
$k = 1/4\pi\epsilon_0$	8.99×10^9	N·m^2/C^2
$k_m = \mu_0/4\pi$	10^{-7}	T·m/A
c	3.00×10^8	m/s
k_B	1.38×10^{-23}	J/K
N_A	6.02×10^{23}	(mol)$^{-1}$
$R = N_A k_B$	8.31	J/(mol·K)
σ	5.67×10^{-8}	J/(s·m^2·K^4)
e	1.602×10^{-19}	C
1 eV	1.602×10^{-19}	J
h	6.63×10^{-34}	J·s = 4.14×10^{-15} eV·s
m_e	9.109×10^{-31}	kg = 0.511 MeV/c^2

Extra reading

- Stefan Ivanov's handouts
- David Thong lectures in General Relativity